Reflections on AI and Quantitative History

HY5419 · Seminar 7

At present, I mostly use generative AI to find and collect literature and data for my research. It is also a useful tool for developing ideas. While writing my research proposal, I found that my description of the topic was not precise enough. So I sent my initial ideas to AI and asked it to question me as someone with no background in history. This helped me understand my own project better. I also think generative AI is a useful learning tool. In the Dunhuang project, I use it to show me how to use digital tools step by step. However, when I read Oberbichler and Petz's distinction between AI as a supporting tool and as a research method, I returned to questions I had been thinking about while preparing our group presentation and during the discussions that followed: Why choose Gephi? What role did it actually play in the Dunhuang project? Is it irreplaceable?[1] I first need to understand what a tool is for and what it can do before deciding whether it suits my research question. I am not against new technology. I also enjoy trying it in my research and seeing the different perspectives and conclusions it can lead to. But I do not think historical research has to change its methods every time tools are updated. I should keep my own research perspective and judgment.

For this reason, I find the paper's recommendation of 'minimal computing' somewhat difficult to understand, especially its advice to use models that are as small as possible. My first reaction was to wonder whether this also reflected some caution about new technology. I can understand the authors' concern about environmental costs and their condition that the research should remain effective.[1] But if a research question has not yet taken shape, how can we know in advance how large a model we need? I worry that restricting our tools too early may also limit our choice of sources and ways of thinking. I have reservations about open-source models too. Some research depends on unique historical sources and work built up over a long period. If researchers share the results of their model training and their experience of using these tools, the whole discipline will benefit. But how can they protect their own contributions and the distinctive features of their work? Using an open-source model does not mean making one's own materials public. Yet if researchers go further and share what they have built up, will their individual efforts be recognised? The costs and restrictions of closed-source services, on the other hand, may make participation harder for researchers with fewer resources. The relationship between openness and originality does not seem easy to manage.

Anderson's discussion of quantitative history reminded me of a recent change in my thinking while organising the literature for my project on Chinese medicines.[2] After reading records of everyday life and commercial advertising, I began to think that reforms concerning medicines were not driven entirely by the government from the top down. I even suspect that merchants and residents were the main drivers of the reforms. This idea still needs to be tested,

but it came from sources about these 'ordinary people'. This is also what attracts me to quantitative methods: once scattered records are organised, they can be examined together, allowing new historical questions to emerge. However, appearing more often in the records does not directly mean that a group had greater influence. We still need to return to the individual records to understand this. This also leaves me unsure whether the question should come first or whether I should read the sources first. When faced with a large database, researchers should certainly know what they are looking for. But my Chinese medicines project shows that reading sources can also help shape the question. I therefore do not want the variables already in a database to determine what I can ask. Quantitative history also shares many methods with sociology and economics. I would still like to discuss whether it needs its own research institutions and training system. Understanding the context of historical sources and change over time requires specific training. But would establishing a separate discipline achieve this, or would it mean spending resources on the same work again?

While reading Jaillant and Rees, I still did not think that better technical capabilities would necessarily make archives more open.[3] AI can make it more efficient to organise information and identify sensitive content. But governments and archive owners ultimately decide which records are worth preserving and which can be made public. The authors also discuss these barriers. What concerns me more is how much shared ethics can constrain people's actions when their interests differ. My undergraduate dissertation was on 'regulatory sandboxes', which also makes me think about the gap between technical ideas and their use in practice. Developing rules requires practical experience, not just agreements. The thirty people interviewed in the article can reveal concerns, but their views cannot tell us what happens in different regions. We also need to consider how to protect privacy and who should take responsibility once records are made accessible.

The Valley of the Shadow database provided by our instructor raised more specific questions for me about how sources are classified. It compares Augusta County in Virginia with Franklin County in Pennsylvania. Users can view each county separately or use pages that compare them side by side, which makes the differences easier to see.[4] The categories help me find sources. But if they do not match my research question, could they instead hide some records? This selection also took place before digitisation: the 1860 slaveowners census recorded the names of slaveowners but not those of enslaved people.[4] Turning archives into a database therefore does not mean that everyone in them becomes easier to see. These statistics are useful for comparing the population and economic conditions of the two counties. Using the 1860 and 1870 records together also allows us to ask questions about changes before and after the war. However, the property figures for the two years were not measured on the same basis, so the difference cannot simply be treated as losses caused by the war. County-level figures also cannot tell me how an individual experienced the war.[4] I hope the website can make it easier for

readers to find the reasons behind its categories, the sources of its figures and information about missing data. Readers should also be able to filter records according to their own questions and export them in common formats, so they do not have to depend on one particular tool. Since comparisons between the two counties already exist, bringing together data from different years that use the same definitions, and linking them to the relevant original records, would make comparison and further questioning easier.

Historians should keep an open mind. They should not simply accept and use digital tools passively. More importantly, they should take an active part in improving them and tell technical staff about the problems they encounter. But this returns to my earlier concern: are researchers willing to share their processes and findings? How can this sharing help others without overlooking the work of the individual researcher?

[1] Oberbichler, Sarah, and Cindarella Petz. Working Paper: Implementing Generative AI in the Historical Studies. IEG, February 2025, pp. 3, 9, 15–16.

[2] Anderson, Margo. "Quantitative History." In The SAGE Handbook of Social Science Methodology, edited by William Outhwaite and Stephen P. Turner. SAGE, 2007.

[3] Jaillant, Lise, and Arran Rees. "Applying AI to Digital Archives: Trust, Collaboration and Shared Professional Ethics." Digital Scholarship in the Humanities 38, no. 2 (2023): 571–585.

[4] The Valley of the Shadow. 1860 comparison;1860 wealth table;1870 comparison;1870 property and taxation table;Census documentation. Accessed 26 September 2026.

Statement: This text was originally written in Chinese and translated using the AI tool ChatGPT.